Internal Energy, Specific Heat and Correlation Function of the Bond-Random Ising model

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Intro

$$Z(T,K) = \sum_{\{T\}} e^{\sum_{\langle i,j \rangle} T_{ij} K \sigma_i \sigma_j}$$

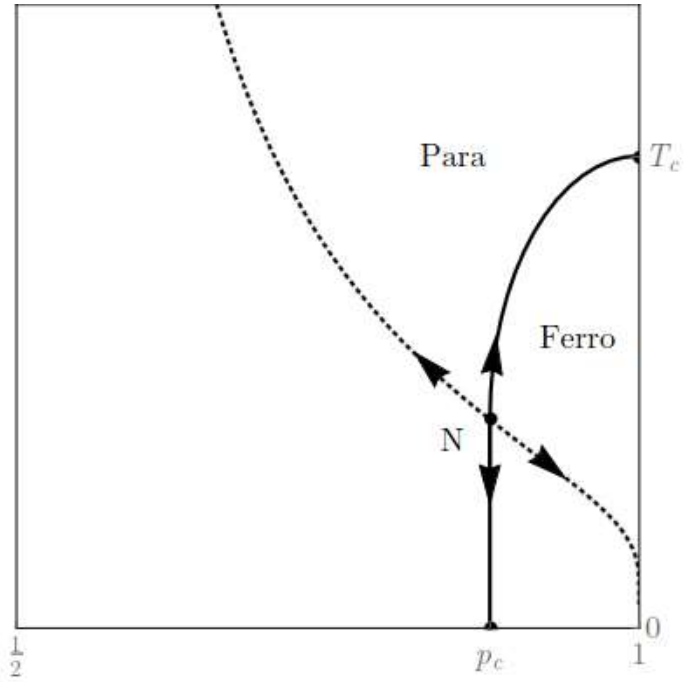
Tij is +1 with probability p and -1 with probability 1-p , also in 2 level system form,

$$e^{-2K_p} = \frac{1-p}{p}$$

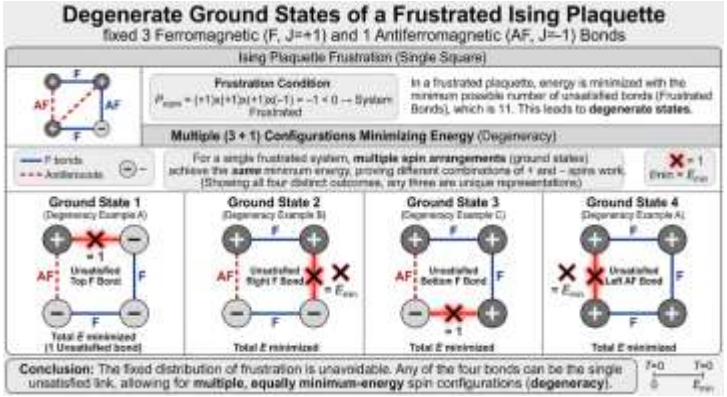
p=0, Purely paramagnetic
p=1 Ising model, solved by Onsager, Tc= 1/(sqrt(2)+1)

At T=0, the problem of minimizing energy has exponentially many different solutions, due to “frustration”

For p<½, the system is exact replica, but antiferromagnetic



Spin Glass and Frustration



From The picture, Each frustration creates 4

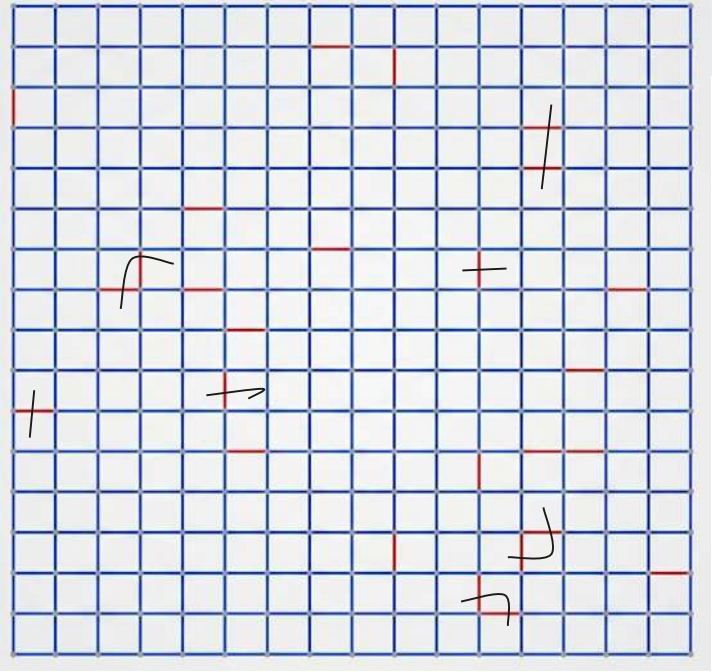
Energy minima, if (1-p)N are frustrated,

O(4^(1-p)*N) that is, exponentially many

Minima exist, such a phase is called a

spin glass phase. In 2D, it can exist only at T=0. For p>½, the antibonds percolate. the boundary of a domain wall if it aligns along a length L antibond chain, which exists for all L, then the energy needed to make that domain wall is O(1), so by Pierl's argument you have no spin glass, as at finite temperature exponential number of energy minima do not exist. In 3D and higher, there is no 2D sheet percolation, only line percolation, so energy of creating a domain is O(L^2) and pierl's argument applies to give us a spin glass phase at finite temperature too.

Entropy of Frustration



Take W = product of T along the edges of each plaqette.

The configuration W is the “frustration”.

$$P(\tau) = \prod_{\langle ij \rangle} \frac{e^{K_p \tau_{ij}}}{2\cosh K_p} = \frac{e^{K_p \sum_{\langle ij \rangle} \tau_{ij}}}{(2\cosh K_p)^{N_b}}$$

Upon transforming Tij -> Tij*Si*Sj using any

Field S on the lattice vertices, you get the

Same W. Thus, probability of frustration W is-

$$P(W) = \sum_{\{\sigma_i\}} P\left(\{\tau_{ij}^{(0)}(W)\sigma_i\sigma_j\}\right) = \frac{1}{(2\cosh K_p)^{N_b}} \sum_{\{\sigma_i\}} \exp\left(K_p \sum_{\langle ij \rangle} \tau_{ij}^{(0)}(W)\sigma_i\sigma_j\right)$$

Phase transition - Plotting W, they come in pairs, and a phase transition is when a pair of these plaqettes go to infinity

Calculating Free energy

$$P(T_{ij}) = \frac{e^{K_p T_{ij}}}{2\cosh K_p}, \quad \text{where } e^{-2K_p} = \frac{1-p}{p}$$

Gauge Transform

$$T_{ij} \longrightarrow T'_{ij} = T_{ij} S_i S_j$$

$$\sigma'_i \sigma'_j = (\sigma_i S_i)(\sigma_j S_j) = (\sigma_i \sigma_j)(S_i S_j)$$

$$Z(\{T_{ij}\}, K) = \sum_{\{\sigma_i\}} \exp\left(K \sum_{\langle ij \rangle} T_{ij} \sigma_i \sigma_j \right)$$

Since Z(W,K) only, average over all T's giving same W

$$-\beta[F] = \frac{1}{2^N} \sum_{\{S_i\}} \sum_{\{T_{ij}\}} P(\{T_{ij} S_i S_j\}) \ln Z(\{T_{ij} S_i S_j\}, K)$$

$$\sum_{\{S_i\}} P(\{T_{ij} S_i S_j\}) = \frac{1}{(2\cosh K_p)^{N_b}} \sum_{\{S_i\}} \exp\left(K_p \sum_{\langle ij \rangle} T_{ij} S_i S_j \right)$$

$$-\beta[F] = \frac{1}{2^N (2\cosh K_p)^{N_b}} \sum_{\{T\}} Z(\{T\}, K_p) \ln Z(\{T\}, K_p)$$

Internal Energy along Nishimori Line

$$[E] = -\frac{J}{2^N (2\cosh K_p)^{N_b}} \sum_{\{T\}} Z(\{T\}, K) \frac{\partial \ln Z(\{T\}, K)}{\partial K} \quad [E] = -\frac{J}{2^N (2\cosh K_p)^{N_b}} \sum_{\{T\}} \frac{\partial Z(\{T\}, K_p)}{\partial K}$$

$$\sum_{\{T_{ij}\}} Z(\{T_{ij}\}, K) = \sum_{\{T_{ij}\}} \sum_{\{\sigma_i\}} \exp\left(K \sum_{\langle ij \rangle} T_{ij} \sigma_i \sigma_j \right) = \sum_{\{\sigma_i\}} \prod_{\langle ij \rangle} \left(\sum_{T_{ij}=\pm 1} e^{K T_{ij} \sigma_i \sigma_j} \right)$$

$$\sum_{T_{ij}=\pm 1} e^{K T_{ij} \sigma_i \sigma_j} = e^K + e^{-K} = 2\cosh K$$

$$\sum_{\{T_{ij}\}} Z(\{T_{ij}\}, K) = \sum_{\{\sigma_i\}} (2\cosh K)^{N_b} = 2^N (2\cosh K)^{N_b}$$

$$\frac{\partial}{\partial K} \left[2^N (2\cosh K)^{N_b} \right] = 2^N N_b (2\cosh K)^{N_b - 1} (2\sinh K) = 2^N (2\cosh K)^{N_b} \cdot N_b \tanh K$$

$$[E] = -N_b J \tanh K_p$$

Heat Capacity

$$\langle E^2 \rangle = \frac{J^2}{Z(\{T_{ij}\}, K)} \frac{\partial^2 Z(\{T_{ij}\}, K)}{\partial K^2}$$

$$[\langle E^2 \rangle] = \frac{J^2}{2^N (2\cosh K_p)^{N_b}} \sum_{\{T_{ij}\}} \frac{\partial^2 Z(\{T_{ij}\}, K_p)}{\partial K^2}$$

$$\sum_{\{T_{ij}\}} Z(\{T_{ij}\}, K) = 2^N (2\cosh K)^{N_b}$$

$$\sum_{\{T_{ij}\}} \frac{\partial^2 Z(\{T_{ij}\}, K)}{\partial K^2} = \frac{\partial^2}{\partial K^2} \sum_{\{T_{ij}\}} Z(\{T_{ij}\}, K)$$

$$[\langle E^2 \rangle] = \frac{J^2}{2^N (2\cosh K_p)^{N_b}} \left[2^N (2\cosh K_p)^{N_b} \left(N_b^2 \tanh^2 K_p + N_b \mathrm{sech}^2 K_p\right)\right]$$

$$[\langle E^2 \rangle] = J^2 \left(N_b^2 \tanh^2 K_p + N_b \mathrm{sech}^2 K_p\right)$$

$$[C] \leq \frac{K^2}{J^2} \left([\langle E^2 \rangle] - [\langle E \rangle]^2\right) = \frac{K_p^2}{J^2} \left(N_b J^2 \mathrm{sech}^2 K_p\right) = N_b K_p^2 \mathrm{sech}^2 K_p$$

As heat capacity is finite on Nishimori line, there is no energy needed for phase transition at the multicritical point. This proves the phase transition is a percolation, not a “boiling”

Correlation Functions

$$\langle S_i S_j \rangle_K = \frac{1}{Z(\{T_{ij}\}, K)} \sum_{\{\sigma_k\}} \sigma_i \sigma_j \exp\left(K \sum_{\langle lm \rangle} T_{lm} \sigma_l \sigma_m \right)$$

$$[\langle S_i S_j \rangle_K] = \sum_{\{T_{ij}\}} P(\{T_{ij}\}) \langle S_i S_j \rangle_K$$

Gauge Transform

$$T_{lm} \longrightarrow T_{lm} S_l S_m$$

$$[\langle S_i S_j \rangle_K] = \frac{1}{2^N (2 \cosh K_p)^{N_b}} \sum_{\{S_k\}} \sum_{\{T_{lm}\}} \exp\left(K_p \sum_{\langle lm \rangle} T_{lm} S_l S_m \right) \langle S_i S_j \rangle_K (\{T_{lm} S_l S_m\})$$

$$\langle S_i S_j \rangle_K (\{T_{lm} S_l S_m\}) = \frac{\sum_{\{\sigma_k\}} \sigma_i \sigma_j \exp\left(K \sum_{\langle lm \rangle} T_{lm} S_l S_m \sigma_l \sigma_m \right)}{Z(\{T_{lm} S_l S_m\}, K)}$$

$$\langle S_i S_j \rangle_K (\{T_{lm} S_l S_m\}) = S_i S_j \langle S_i S_j \rangle_K (\{T_{lm}\})$$

Correlation Functions

$$[\langle S_i S_j \rangle_K] = \frac{1}{2^N (2\cosh K_p)^{N_b}} \sum_{\{S_k\}} \sum_{\{T_{lm}\}} \exp\left(K_p \sum_{\langle lm \rangle} T_{lm} S_l S_m \right) \langle S_i S_j \rangle_K (\{T_{lm} S_l S_m\})$$

$$\sum_{\{S_k\}} S_i S_j \exp\left(K_p \sum_{\langle lm \rangle} T_{lm} S_l S_m \right) = Z(\{T_{lm}\}, K_p) \cdot \langle S_i S_j \rangle_{K_p}$$

$$[\langle S_i S_j \rangle_K] = \frac{1}{2^N (2\cosh K_p)^{N_b}} \sum_{\{T_{lm}\}} Z(\{T_{lm}\}, K_p) \langle S_i S_j \rangle_{K_p} \langle S_i S_j \rangle_K$$

This is just sum over all P(T)'s

$$[\langle S_i S_j \rangle_K] = [\langle S_i S_j \rangle_{K_p} \langle S_i S_j \rangle_K]$$

Correlations and multicriticality

$$|[\langle S_i S_j \rangle_K]| \leq [\langle S_i S_j \rangle_{K_p}]$$

As spin correlations are less than 1

So, if it is paramagnetic on Nishimori line for a given p, with the correlations going to 0 for large |i-j|, it is paramagnetic for all T as well. This implies the phase boundary is vertical or curves inward below the “multicritical point”.

Also, with K=Kp, the average magnetisation m^2 = q^2 =[<Si*Sj>^2]. A spin glass is when q>0 and m=0, so along Nishimori line it is impossible, and naturally impossible for larger temperatures.

For T<Tc and p<pc, ferromagnetism is impossible, but for d>2, spin glass phase exists.

Thank You